

Knowledge Organiser — 1.4 Data Types, Data Structures & Boolean Algebra

OCR A Level Computer Science (H446) · Component 01 · 1.4.1 Data types · 1.4.2 Data structures · 1.4.3 Boolean algebra

1.4.1 Data Types

Primitive types

Type	Holds	Example
Integer	Whole number (no fraction)	42, -7
Real / Float	Number with fractional part	3.14
Char	Single character	'A'
String	Sequence of characters	"Hi"
Boolean	True / False	True

Binary place values (8-bit unsigned, range 0–255)

128	64	32	16	8	4	2	1

Hex = base 16 (0–9, A=10...F=15). **1 hex digit = 4 bits (nibble)**; 2 hex = 1 byte.

Conversion methods

- **Denary** → **binary**: subtract largest place value that fits, repeat (or $\div 2$, read remainders bottom-up).
- **Binary** → **denary**: add place values where bit = 1.
- **Binary** → **hex**: split into nibbles **from the right** (pad left with 0s), convert each.
- **Hex** → **binary**: expand each digit to 4 bits. **Hex** → **denary**: $(\text{digit}_1 \times 16) + \text{digit}_0$.
- *Why hex?* Compact, lossless, error-free shorthand for binary (addresses, colours, machine code).

Two's complement (signed) — 8-bit range –128 to +127

-128	64	32	16	8	4	2	1

- MSB has a **negative** place value; MSB = 1 $\Rightarrow$ negative.
- **Negate (find $-x$)**: invert **ALL** bits, then **ADD 1**. e.g. +5 00000101 $\rightarrow$ invert 11111010 $\rightarrow$ +1 $\rightarrow$ 11111011 = -5.

Binary addition rules

$0+0=0$ · $1+0=1$ · $1+1=10$ (write 0 carry 1) · $1+1+1=11$ (write 1 carry 1) - **Overflow**: carry beyond available bits / two same-sign numbers give a wrong-sign result.

Binary subtraction

- $A - B = A + (-B)$ — add the two's complement of B; **discard final carry** out of MSB.

Binary shifts

Shift	Effect
Logical left $\times n$	fill 0s from right $\rightarrow \times 2^n$
Logical right $\times n$	fill 0s from left $\rightarrow \div 2^n$ (integer; lost bits)
Arithmetic right $\times n$	copies sign bit in from left $\rightarrow \div 2^n$ for signed numbers
- Bits shifted off the end are lost ; left shift can overflow.	

Bitwise ops

AND = mask/keep bits · **OR** = set bits to 1 · **XOR** = toggle/differ · **NOT** = invert.

Floating point — value = mantissa $\times 2^{\text{exponent}}$

- **Mantissa** = significant figures $\rightarrow$ **precision**. **Exponent** = position of point $\rightarrow$ **range**.
- Both stored in **two's complement** (signed). Point assumed after first (sign) bit of mantissa.
- **Normalisation rule**: positive starts **0.1...** ; negative starts **1.0...** (removes wasted leading bits $\rightarrow$ max precision).
- Moving point **right** $\Rightarrow$ **exponent decreases**; moving **left** $\Rightarrow$ **exponent increases** (value unchanged).
- **Trade-off (fixed total bits)**: more mantissa = more precision, less range; more exponent = more range, less precision.

Character sets

- **ASCII**: 7-bit = 128 chars ('A' =65, 'a' =97, '0' =48); extended 8-bit = 256.
 - **Unicode**: more bits (UTF-8 = 1–4 bytes) $\rightarrow$ ~1.1M code points, all languages + emoji. **ASCII is a subset** (first 128 codes) $\Rightarrow$ backward-compatible.
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1.4.2 Data Structures

Static vs dynamic

	Static	Dynamic
Size	Fixed at declaration	Grows/shrinks at run time
Memory	Contiguous block up front	Allocated from heap as needed
Trade-off	Fast access, may waste/run out	Flexible, pointer overhead
Example	Array	Linked list

Structure	Key facts & operations	Static/Dynamic
Array (1D/2D/3D)	Fixed size, same type , indexed (0-based), $O(1)$ access. 2D <code>grid[r][c]</code> , 3D <code>cube[l][r][c]</code>	Static
Record	Named fields of different types; <code>student.name</code>	Static
List	Ordered, mutable , mixed types, resizable; append/insert/remove/index	Dynamic
Tuple	Ordered, immutable (cannot change); e.g. (x, y)	Static
Linked list	Nodes = data + pointer to next; head pointer, last = null. Easy insert/delete (repoint); no random access , must traverse	Dynamic
Stack (LIFO)	push/pop/peek at top via stack pointer. Overflow/underflow. Uses: call stack, undo, backtracking	Either
Queue (FIFO)	enqueue at rear , dequeue at front . Circular: <code>rear=(rear+1) MOD size</code> reuses space. Priority: highest priority first. Uses: spooling, buffers, scheduling	Either
Graph	Vertices + edges; directed/undirected/weighted. Adjacency matrix (dense, $O(1)$ lookup) vs list (sparse, memory-efficient)	Either
Tree	Connected, acyclic; root, parent/child, leaf. Binary tree ≤ 2 children	Either
BST	Binary tree where left subtree < node < right subtree ; $O(\log n)$ search when balanced. Insert: go left if smaller, right if larger	Either
Hash table	Key $\rightarrow$ index via hash function (e.g. <code>key MOD size</code>), $\sim O(1)$. Collision = 2 keys same index $\rightarrow$ linear probing or chaining. Load factor $\uparrow$ = slower	Either

Tree traversals: Pre-order (root,L,R) prefix/copy · **In-order (L,root,R) = BST ascending** · Post-order (L,R,root) postfix/delete.

1.4.3 Boolean Algebra

Logic gates truth table

A	B	AND	OR	XOR	NAND	NOR
0	0	0	0	0	1	1
0	1	0	1	1	1	0
1	0	0	1	1	1	0
1	1	1	1	0	0	0

NOT $\neg A$: inverts. **AND** $A \cdot B$ · **OR** $A+B$ · **XOR** $A \oplus B$ · **NAND** $\neg(A \cdot B)$ · **NOR** $\neg(A+B)$ · **NAND** & **NOR** are **universal**.

Boolean identities

Law	Form
Identity	$A \cdot 1 = A \cdot A+0 = A$
Null	$A \cdot 0 = 0 \cdot A+1 = 1$
Idempotent	$A \cdot A = A \cdot A+A = A$
Complement	$A \cdot \bar{A} = 0 \cdot A+\bar{A} = 1$
Double negation	$\neg(\neg A) = A$
Commutative	$A \cdot B = B \cdot A \cdot A+B = B+A$
Associative	$A \cdot (B \cdot C) = (A \cdot B) \cdot C \cdot A+(B+C) = (A+B)+C$
Distributive	$A \cdot (B+C) = A \cdot B + A \cdot C \cdot A+(B \cdot C) = (A+B) \cdot (A+C)$
Absorption	$A + A \cdot B = A \cdot A \cdot (A+B) = A$
De Morgan's	$\neg(A \cdot B) = \neg A + \neg B \cdot \neg(A+B) = \neg A \cdot \neg B$

De Morgan's: "break the bar, change the sign" — negate **each** variable AND swap AND $\leftrightarrow$ OR. **K-maps**: group adjacent 1s in powers of 2 (1,2,4,8...), as large as possible, may wrap/overlap $\rightarrow$ minimal sum-of-products.

Adders

Half adder (2 bits, no carry-in): - **Sum** = $A \oplus B$ · **Carry** = $A \cdot B$

Full adder (A, B, Cin) = two half adders + OR; chain $\rightarrow$ ripple-carry adder: - **Sum** = $A \oplus B \oplus \text{Cin}$ - **Cout** = $(A \cdot B) + (\text{Cin} \cdot (A \oplus B))$

D-type flip-flop

- **Edge-triggered 1-bit memory** (sequential logic). On active **clock edge**, stores input **D** and presents it at **Q** until next edge; synchronises data to the clock.
 - Uses: registers, counters, memory cells. (Combinational = output from inputs only; sequential = inputs + stored state.)
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Must-know definitions

Term	Definition
Two's complement	Signed-integer rep where MSB is a negative place value; negate = invert all bits + 1
Overflow	Result too large for the available bits
Mantissa	Significant-figures part of a float; sets precision
Exponent	Power-of-two part of a float; sets range
Normalisation	Float adjusted so mantissa begins with first significant bit (0.1... +ve / 1.0... -ve) for max precision
Logical shift	Fills with 0s; $\times/\div$ unsigned by powers of 2
Arithmetic shift	Right shift preserving sign bit; $\div$ signed by powers of 2
Character set	Mapping of characters to unique binary codes (ASCII, Unicode)
Array	Ordered, fixed-size, same-type collection accessed by index
Record	Collection of named fields, possibly different types
Tuple	Ordered, immutable sequence
Linked list	Dynamic list of nodes, each = data + pointer to next
Stack / Queue	LIFO (push/pop at top) / FIFO (enqueue rear, dequeue front)
BST	Binary tree where left subtree < node < right subtree
Hash table	Keys $\rightarrow$ indices via hash function for fast lookup; collision = 2 keys same index
Half / Full adder	Adds 2 bits $\rightarrow$ Sum (XOR) + Carry (AND) / adds 3 bits (incl. Cin) $\rightarrow$ Sum + Cout
D-type flip-flop	Edge-triggered 1-bit memory storing D on a clock edge

Top exam reminders (discriminators)

- **Two's complement negation:** invert **all** bits **then +1** — don't forget the +1; in subtraction **discard the final carry**.
- **Shifts:** always state **direction AND effect** ($\times$ or $\div 2^n$); bits shifted off are **lost**; arithmetic right keeps the **sign bit**.
- **Normalisation:** +ve starts 0.1, -ve starts 1.0; point **right** $\Rightarrow$ **exponent down**, left $\Rightarrow$ up. **More mantissa = precision, more exponent = range** (don't swap).
- **De Morgan's:** change the operator **and** negate **each** variable (not just one); tidy with double negation.
- **Structures:** BST $\neq$ any binary tree (needs ordering property); name LIFO vs FIFO ends correctly; half adder has **no carry-in**, full adder adds three bits.